The documentation and process conversion measures necessary to comply with this document shall be completed by 16 June 2021.

INCH-POUND

MIL-PRF-19500/159R
w/AMENDMENT 2
16 March 2021
SUPERSEDING
MIL-PRF-19500/159R
w/AMENDMENT 1
9 August 2019

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE, TEMPERATURE COMPENSATED, TYPES 1N821-1, 1N823-1, 1N825-1, 1N827-1, AND 1N829-1, ENCAPSULATED (AXIAL LEADED AND SURFACE MOUNT PACKAGE) AND UN-ENCAPSULATED, RADIATION HARDENED (TOTAL DOSE ONLY), QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC AND JANKC;

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 6.2 volts ±5 percent, silicon, temperature compensated, voltage-reference diodes. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance (JANHC and JANKC) are provided for each un-encapsulated device type. Seven levels of radiation hardened assurance (total dose only) are provided for quality levels JANTXV, JANS, JANHC and JANKC devices as specified in MIL-PRF-19500.

1.2 Physical dimensions.

1.2.1 Package outlines. The device package outlines are as follows: Axial leaded packages DO-204AA or DO-204AH (similar to DO-7 and DO-35) in accordance with figure 1 and surface mount version metal electrode leadless face (MELF) (similar to DO-213AA) in accordance with figure 2.

1.2.2 Un-encapsulated die dimensions and topography. The dimensions and topography for un-encapsulated die are in accordance with figure 3 (JANHC and JANKC), figure 4 (JANHCB and JANKCB) figure 5 (JANHCC and JANKCC), and figure 6 (JANHCD and JANKCD).

1.3 Maximum ratings. Unless otherwise specified T_A = +25°C.

P_T (1)	T_{STG} and T_J	I_{ZM} (1)	Power derating above T_A = +25°C
mW	°C	mA dc	mW/°C
500	−55 to +175	70	3.33

(1) To guarantee voltage temperature stability, it is necessary to maintain the proper I_Z = 7.5 mA dc.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Type	ΔV_Z (1) (voltage -temperature stability)	Z_Z (1)	V_Z (1)		I_R V_R = 3.0 V
			Min	Max	
	mV dc	ohms	volts	volts	µA
1N821-1, 1N821UR-1	96	15	5.89	6.51	2.0
1N823-1, 1N823UR-1	48	15	5.89	6.51	2.0
1N825-1, 1N825UR-1	19	15	5.89	6.51	2.0
1N827-1, 1N827UR-1	9	15	5.89	6.51	2.0
1N829-1, 1N829UR-1	5	15	5.89	6.51	2.0

(1) To guarantee voltage temperature stability, it is necessary to maintain the proper I_Z of 7.5 mA dc.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Quality level designators for un-encapsulated devices (die). The quality level designators for un-encapsulated devices that are applicable for this specification sheet are "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "821", "823", "825", "827" and "829".

1.5.3 Suffix symbols. The following suffix symbol(s) are incorporated into the PINs for this specification sheet.

1.5.3.1 Package designators. The suffix symbols (or lack thereof) that designate the package outline for the devices covered by this specification sheet are as follows.

	A blank designator identifies that the package is a DO-204AA or DO-204AH (formerly DO-7or DO-35) in accordance with figure 1.
* UR	A "UR" designator identifies that the package is a DO-213AA surface mount package in accordance with figure 2.

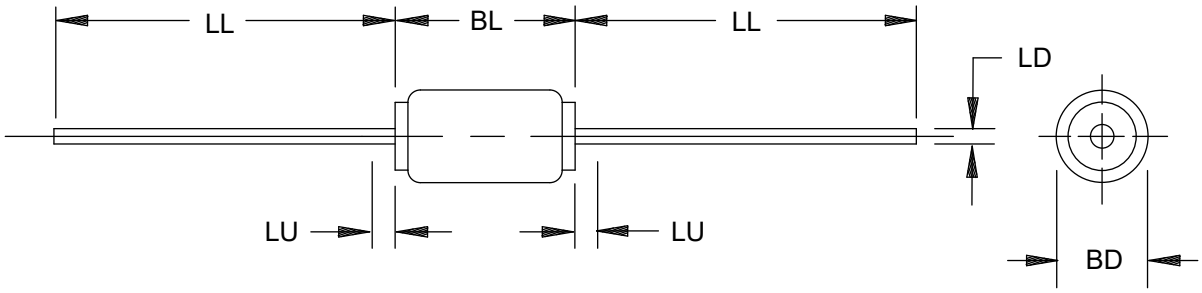
1.5.3.2 Metallurgical bond. The last suffix symbol "-1" designates the diode construction (see 3.4.2).

1.5.4 Radiation hardness assurance (RHA) designator. For quality levels JANTXV, JANS, JANHC and JANKC, the RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "L", "R", "F", "G", and "H".

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.6 Die identifiers for un-encapsulated devices. The manufacturer die identifiers that are applicable for this specification sheet are "A", "B", "C" and "D" (see figures 3, 4, 5, 6 and 6.5).

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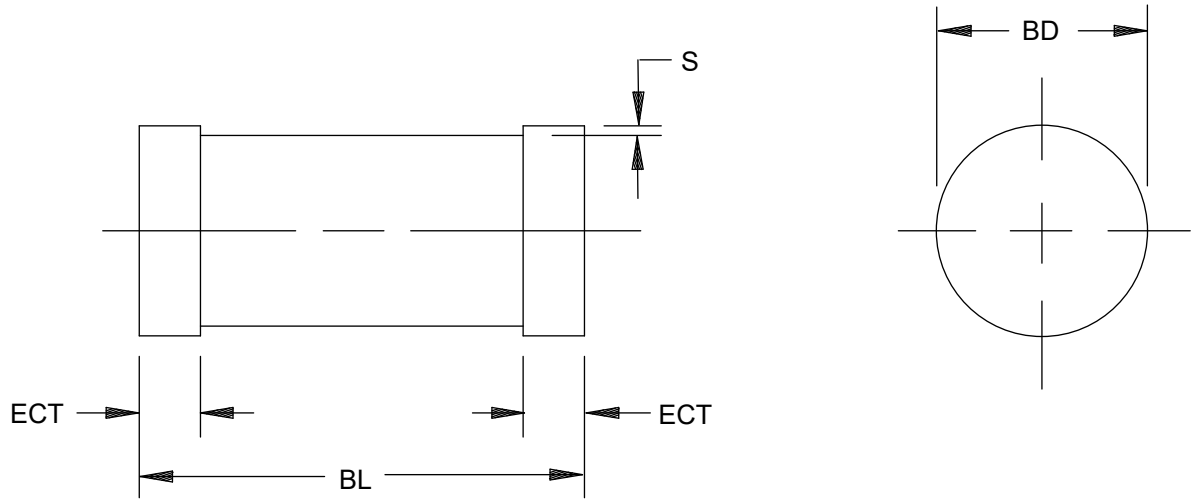


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.060	.107	1.52	2.72	2
BL	.120	.300	3.05	7.62	2
LD	.018	.023	0.46	0.58	
LL	1.000	1.500	25.40	38.10	
LU		.050		1.27	3

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Package contour optional within BD and length BL. Heat slugs, if any shall be included within this cylinder but shall not be subject to minimum limit of BD.
3. Within this zone, lead diameter may vary to allow for lead finishes and irregularities, other than heat slugs.
* 4. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions of DO-204AA or DO-204AH packages (1N821-1 thru 1N829-1).

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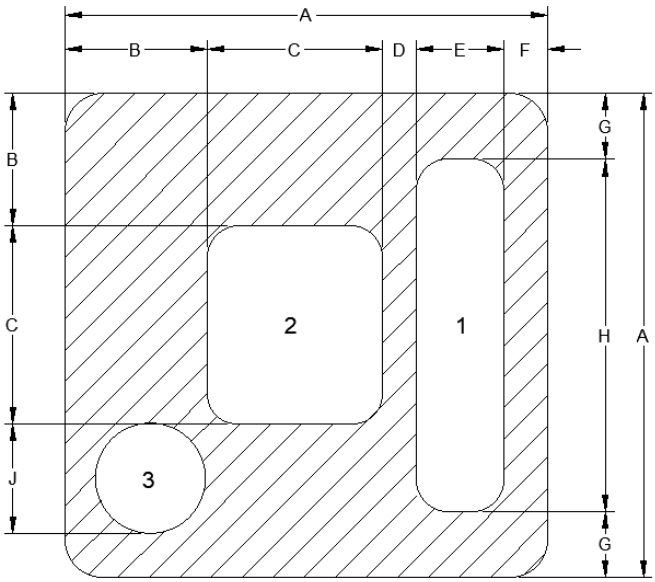
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.063	.067	1.60	1.70
BL	.130	.146	3.30	3.71
ECT	.016	.022	0.41	0.56
S	.001 min		0.03 min	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
* 2. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.
* 3. Dimension S is optional however the glass body diameter shall not exceed endcap diameter.

* FIGURE 2. Physical dimensions of DO-213AA package (1N821UR-1 through 1N829UR-1).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.0280	.0320	.711	.813
B	.0080	.0100	.203	.254
C	.0104	.0106	.264	.269
D	.0019	.0021	.048	.053
E	.0054	.0056	.137	.142
F	.0020	.0040	.050	.102
* G	.0030	.0050	.076	.127
* H	.0209	.0211	.531	.536
* J	.0059	.0061	.150	.155



Design data

Circuit layout data:

Backside must be electrically isolated to ensure proper performance.

For Zener operation, cathode must be operated positive with respect to anode.

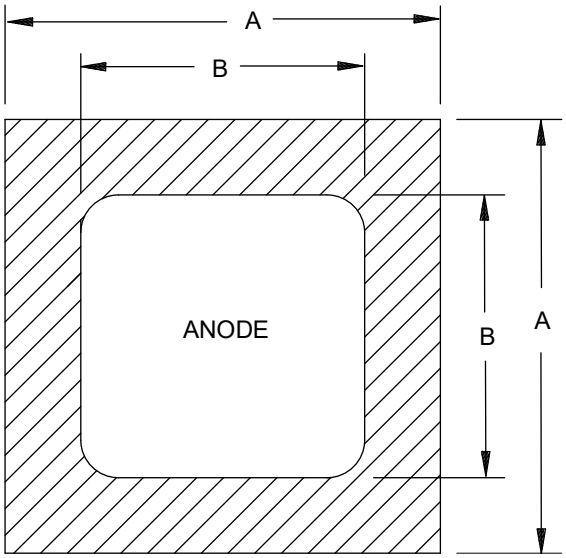
Metallization:	Top:	1 (Cathode)	Al	
		2 (Anode)	Al	
		3 (Test pad)	Al	Test pad is for wire bond evaluation only. No electrical contact is made with test pad.
	Back:		Au	
	Al thickness		25,000Å minimum	
	Au thickness		4,000Å minimum	

Die thickness: .010 inch (0.25 mm) ±0.002 inch (±0.05 mm).

NOTES:
1. Dimensions are in inches unless otherwise indicated. Millimeters are given for general information only.
* 2. In accordance with ASME Y14.5 diameters are equivalent to ϕx symbology.

FIGURE 3. JANHC and JANKC (A-version) die dimensions.

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BACKSIDE IS CATHODE

Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.024	.037	0.61	0.94
B	.014	.029	0.36	0.74

Design data

Circuit layout data:

For zener operation, cathode must be operated positive with respect to anode.

Metallization:	Top: (Anode)	Al
	Back: (Cathode)	Au
	Al thickness	40,000Å minimum
	Au thickness	5,000Å minimum

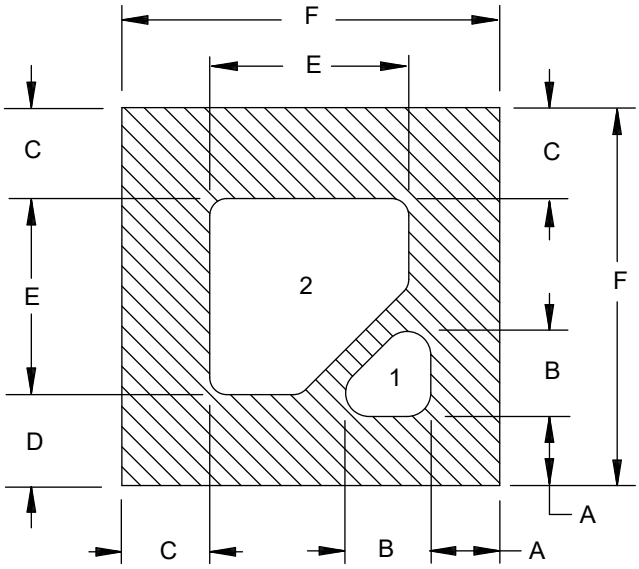
Die thickness: .010 inch (0.25 mm) ±.002 inch (±0.05 mm).

NOTES:
1. Dimensions are in inches. Millimeters are given for general information.
* 2. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 4. JANHC and JANKC (B-version) die dimensions.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.0035	.0065	.088	.165
B	.0050	.0080	.127	.203
C	.0050	.0065	.127	.165
D	.0050	.0065	.127	.165
E	.0150	.0165	.381	.419
F	.0260	.0290	.660	.737



Design data

Circuit layout data:

For zener operation, cathode must be operated positive with respect to anode.

Backside must be electrically isolated.

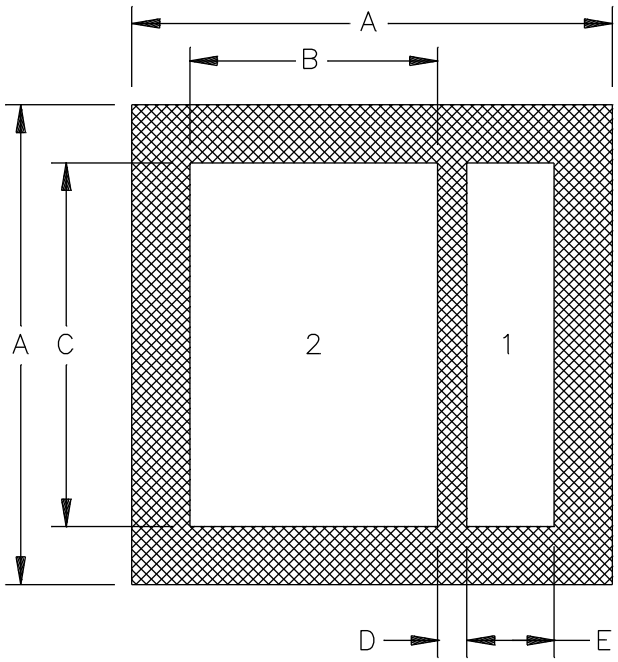
Metallization:	Top:	1 (Cathode)	Al
		2 (Anode)	Al
	Back		Au
	Al thickness		40,000Å minimum
	Au thickness		5,000Å minimum

Die thickness: .010 inch (0.25 mm) ±.002 inch (±0.05 mm).

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
* 2. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 5. JANHC and JANKC (C-version) die dimensions.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.031	.035	0.787	0.889
B	.016	.018	0.406	0.457
C	.024	.026	0.610	0.660
D	.001	.003	0.025	0.076
E	.005	.007	0.127	0.178
F	.008	.012	0.203	0.305

Design data

Metallization:	Top	1 (Cathode)	Al
		2 (Anode)	Al
	Back		Au
	Al thickness		37,500Å minimum
	Au thickness		3,750Å minimum

Die thickness: .010 inch (0.25 mm) ±0.002 inch (±0.05 mm).

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
* 2. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 6. JANHC and JANKC (D-version) die dimensions.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1, 2, 3, 4, 5, and 6 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Encapsulated diode construction. These devices shall be constructed to comply with the dash-one construction requirements of MIL-PRF-19500 utilizing either a category I or II metallurgical bond as defined in MIL-PRF-19500.

3.4.3 JANHC and JANKC construction. JANHC and JANKC construction may differ in die size and bonding pad layout provided the manufacturing technology is identical (example: diffused junction, alloy junction).

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3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Marking of "UR" version devices. For "UR" version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.5.2 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end or alternately with a minimum of three contrasting color dots spaced evenly around the periphery at the cathode end.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.8 RHA requirements. The requirements and test levels for RHA shall be as defined in MIL-PRF-19500 and herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

d. Element evaluation of un-encapsulated die (see 4.6).

4.1.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-PRF-19500 and as specified herein, except that lot accumulation period shall be 3 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and 4.4.5 herein.

4.2.2 Radiation hardened devices. See MIL-PRF-19500 and 4.4.4 herein.

4.2.3 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with MIL-PRF-19500.

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4.3 Screening of encapsulated devices (quality levels JANS, JANTXV and JANTX only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTXV and JANTX level
3b	Not applicable	Not applicable
3c	Not applicable	Not applicable
10	Not applicable	Not applicable
11	Required V_Z, Z_Z	Required V_Z, Z_Z
12	Required see 4.3.1	Required see 4.3.1
13	Required Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading. T_A = +25°C ±2°C, $\Delta V_Z \leq \pm 0.004$ V dc from initial value for 1N821-1, 1N821UR-1, 1N823-1, 1N823UR-1, 1N825-1 1N825UR-1. $\Delta V_Z \leq \pm 0.003$ V dc from initial value for 1N827-1, 1N827UR-1, 1N829-1, 1N829UR-1.	Required Subgroup 2 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading. T_A = +25°C ±2°C, $\Delta V_Z \leq \pm 0.004$ V dc from initial value for 1N821-1, 1N821UR-1, 1N823-1, 1N823UR-1, 1N825-1 1N825UR-1. $\Delta V_Z \leq \pm 0.003$ V dc from initial value for 1N827-1, 1N827UR-1, 1N829-1, 1N829UR-1.

4.3.1 Power burn-in conditions. Power burn-in conditions shall be as follows: I_Z = 7.5 mA dc, ±0.75 mA dc, T_A = +150°C, +5°C, −0°C.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV.

4.4.2.1 Quality level JANS. Group B inspection for quality level JANS shall be conducted in accordance with the conditions specified for subgroup testing in table E–VIA of MIL-PRF-19500 and herein. For purposes of JANS inspection, a single device type shall be defined as devices from a single wafer lot (for each die type used in the construction). The conformance inspection sample shall be selected from the part category with the lowest ΔV_Z rating in the inspection lot.

Subgroup	Method	Conditions
B3	4066	Not applicable.
B4	1037	I_Z = 35 mA dc at T_A = room ambient; t_{on} = t_{off} = 30 seconds minimum for 4,000 cycles. Forced air cooling allowed during off cycle.
B5	1027	I_{ZM} = 70 mA dc for 96 hours. T_A = +75°C, adjust T_A to achieve T_J = +200°C minimum.
B6		Not applicable.

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4.4.2.2 Quality levels JAN, JANTX and JANTXV. Group B inspection for quality levels JAN, JANTX, and JANTXV shall be conducted in accordance with the conditions specified in table E-VIB of MIL-PRF-19500 and herein.

Subgroup	Method	Conditions
B2	1056	Test condition A, 25 cycles.
B2	4066	Not applicable.
B3	1027	I_Z = 7.5 mA dc, ±0.75 mA dc, T_A = +150°C, +5°C, -0°C.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Tension: Test condition A; 10 pounds (4.54 kg) weight, t = 15 ±3 seconds. Lead fatigue: Test condition E. Tension and lead fatigue test is not applicable to surface mount "UR" version device.
C5		Not applicable.
C6	1026	I_Z = 7.5 mA dc, T_A = +100°C minimum (see 4.5.2).
C7		Not applicable.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein. Submitted lots for group D sample inspection must be constructed using one homogeneous wafer lot for the Zener and one wafer lot for the compensating die, as also described in the Design and Construction form 36D (see table II herein) submitted to the qualifying activity.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup tests in table E-IX of MIL-PRF-19500 and table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage-temperature stability. The breakdown voltage of each diode type shall be measured in accordance with 4.5.3 and recorded at each of the specified temperatures. The lowest measured voltage shall be subtracted from the highest measured voltage for each diode. The difference value obtained shall not exceed the specified ΔV_Z for each diode type.

4.5.2 Reference voltage time stability. The breakdown voltage shall be measured in accordance with 4.5.3 prior to life testing at 340 hours, and at the conclusion of the life test as shown in table IV herein. The 340-hour reading shall be compared with the 0-hour reading and the 1,000-hour reading compared with the 340-hour reading. The change in breakdown voltage shall not exceed the limits specified. The test temperature for breakdown voltage shall be the same as the specified ambient life test temperature (see table IV herein).

4.5.3 Reference voltage. The reference voltage test shall be performed in accordance with test method 4022 of MIL-STD-750. The test current shall be applied until thermal equilibrium is attained (15 seconds minimum) prior to reading the reference voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located between .375 inch (9.53 mm) and .500 inch (12.70 mm) inch from the body and the mounting clips shall be maintained at the specified temperature. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

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4.6 Element evaluation of un-encapsulated die. The element evaluation of un-encapsulated die shall be in accordance with appendix G of MIL-PRF-19500.

* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reference voltage	4022	I_Z = 7.5 ±0.01 mA dc (see 4.5.3)	V_Z	5.89	6.51	V dc
Small-signal breakdown impedance	4051	I_Z = 7.5 ±0.01 mA dc, I_{sig} = 0.75 mA ac	Z_Z		15	ohms
Subgroup 3						
Voltage temperature stability	4022	I_Z = 7.5 ±0.01 mA dc T_A ±2°C = −55°C, 0°C, +25°C, +75°C, +100°C (see 4.5.1)	ΔV_Z			
1N821-1, 1N821UR-1					96	mV dc
1N823-1, 1N823UR-1					48	mV dc
1N825-1, 1N825UR-1					19	mV dc
1N827-1, 1N827UR-1					9	mV dc
1N829-1, 1N829UR-1					5	mV dc
Subgroups 4, 5, and 6						
Not applicable						
Subgroup 7						
* Reverse current leakage	4016	DC method Condition A or B; V_R = 3.0 V dc	I_R		2.0	µA

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits 3/		Post-irradiation limits 3/		ΔV_{Z1} 4/		Unit
	Method	Conditions		Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation	1019	I_Z = 7.5 mA dc condition A								
Reference voltage	4022	I_Z = 7.5 mA dc ±0.01 mA dc (see 4.5.3)	V_Z	5.89	6.51	5.89	6.51			V dc
Small-signal breakdown Impedance	4051	I_Z = 7.5 mA dc I_{stg} = 0.75 ac	Z_Z		15		15			ohms
Reverse current leakage	4016	DC method; V_R = 3.0 V dc	I_R		2.0		2.0			µA
Voltage temperature stability	4022	I_Z = 7.5 mA dc ±0.01 mA dc T_A = 25°C ±2°C (see 4.5.1)	ΔV_Z							
1N821-1, 1N821UR-1									±3.0	mV
1N823-1, 1N823UR-1									±3.0	mV
1N825-1, 1N825UR-1									±2.0	mV
1N827-1, 1N827UR-1									±1.5	mV
1N829-1, 1N829UR-1									±1.0	mV

1/ For sampling plan, see MIL-PRF-19500.
2/ Group D qualification may be performed any time prior to lot formation.
3/ The limits listed are for quality levels JANS and JANTXV of RHA levels "M", "D", "L", "R", "F", "G", and "H".
4/ Pre-irradiation to post-irradiation change.

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TABLE III. Group E inspection qualification and requalification (all product assurance levels).

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling	1051	500 cycles.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices, c = 0
Steady state operation life	1038	Condition B, 1,000 hours (see 4.3.1).	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroups 4, 5, 6, and 7			
Not applicable			
Subgroup 8			
Resistance to glass cracking	1057	Step stress to destruction by increasing cycles or up to a maximum of 25 cycles.	45 devices

TABLE IV. Reference voltage time stability.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Reference-voltage time stability	4022	T_A = +100°C ±2°C, I_Z = 7.5 ±0.01 mA dc, (see 4.5.2)	ΔV_Z			
1N821-1, 1N821UR-1		0 to 340 hours			7	mV dc
1N823-1, 1N823UR-1					7	mV dc
1N825-1, 1N825UR-1					7	mV dc
1N827-1, 1N827UR-1					6	mV dc
1N829-1, 1N829UR-1					5	mV dc
1N821-1, 1N821UR-1		340 to 1,000 hours			4	mV dc
1N823-1, 1N823UR-1					4	mV dc
1N825-1, 1N825UR-1					4	mV dc
1N827-1, 1N827UR-1					3	mV dc
1N829-1, 1N829UR-1					3	mV dc

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

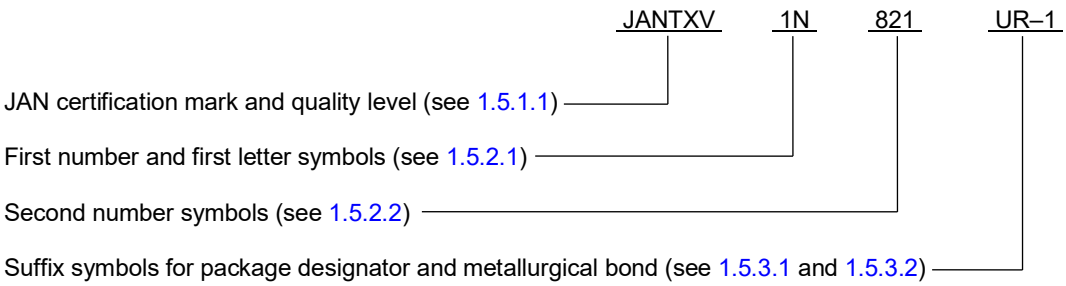
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil . An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

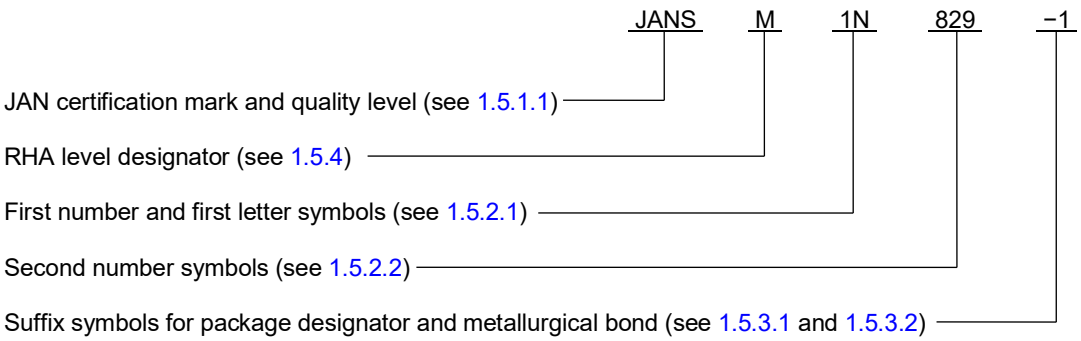
6.4 PIN construction example. The PINs for encapsulated and un-encapsulated devices and are constructed using the following forms.

6.4.1 Non-RHA devices The PINs for encapsulated devices are constructed using the following form.

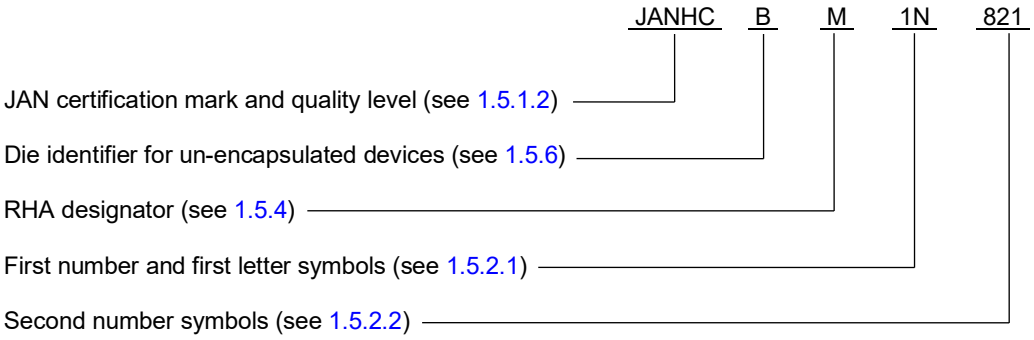


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6.4.2 RHA devices. The PINs for RHA devices are constructed using the following form.



6.4.3 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 Non-RHA devices. The following is a list of possible PINs available for non-RHA devices covered by this specification sheet.

PINs for devices in a axial package			
JAN1N821-1	JANTX1N821-1	JANTXV1N821-1	JANS1N821-1
JAN1N823-1	JANTX1N823-1	JANTXV1N823-1	JANS1N823-1
JAN1N824-1	JANTX1N824-1	JANTXV1N824-1	JANS1N824-1
JAN1N825-1	JANTX1N825-1	JANTXV1N825-1	JANS1N825-1
JAN1N827-1	JANTX1N827-1	JANTXV1N827-1	JANS1N827-1
JAN1N829-1	JANTX1N829-1	JANTXV1N829-1	JANS1N829-1
PINs for devices in a UR package			
JAN1N821UR-1	JANTX1N821UR-1	JANTXV1N821UR-1	JANS1N821UR-1
JAN1N823UR-1	JANTX1N823UR-1	JANTXV1N823UR-1	JANS1N823UR-1
JAN1N824UR-1	JANTX1N824UR-1	JANTXV1N824UR-1	JANS1N824UR-1
JAN1N825UR-1	JANTX1N825UR-1	JANTXV1N825UR-1	JANS1N825UR-1
JAN1N827UR-1	JANTX1N827UR-1	JANTXV1N827UR-1	JANS1N827UR-1
JAN1N829UR-1	JANTX1N829UR-1	JANTXV1N829UR-1	JANS1N829UR-1

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6.5.2 RHA devices. The following is a list of possible PINs available for RHA devices covered by this specification sheet.

PINs for devices in axial and UR packages (1)			
JANTXV#1N821-1	JANS#1N821-1	JANTXV#1N821UR-1	JANS#1N821UR-1
JANTXV#1N823-1	JANS#1N823-1	JANTXV#1N823UR-1	JANS#1N823UR-1
JANTXV#1N824-1	JANS#1N824-1	JANTXV#1N824UR-1	JANS#1N824UR-1
JANTXV#1N825-1	JANS#1N825-1	JANTXV#1N825UR-1	JANS#1N825UR-1
JANTXV#1N827-1	JANS#1N827-1	JANTXV#1N827UR-1	JANS#1N827UR-1
JANTXV#1N829-1	JANS#1N829-1	JANTXV#1N829UR-1	JANS#1N829UR-1

(1) The number sign (#) represents one of seven RHA designators available on this specification sheet ("M", "D", "L", "R", "F", "G", or "H").

6.5.3 List of PINs for non-RHA un-encapsulated die. The following is a list of possible PINs for non-RHA un-encapsulated die covered by this specification sheet.

PINs for types JANHC and JANKC			
JANHCA1N821	JANHCB1N821	JANHCC1N821	JANHCD1N821
JANHCA1N823	JANHCB1N823	JANHCC1N823	JANHCD1N823
JANHCA1N825	JANHCB1N825	JANHCC1N825	JANHCD1N825
JANHCA1N827	JANHCB1N827	JANHCC1N827	JANHCD1N827
JANHCA1N829	JANHCB1N829	JANHCC1N829	JANHCD1N829
JANKCA1N821	JANKCB1N821	JANKCC1N821	JANKCD1N821
JANKCA1N823	JANKCB1N823	JANKCC1N823	JANKCD1N823
JANKCA1N825	JANKCB1N825	JANKCC1N825	JANKCD1N825
JANKCA1N827	JANKCB1N827	JANKCC1N827	JANKCD1N827
JANKCA1N829	JANKCB1N829	JANKCC1N829	JANKCD1N829

6.5.4 List of PINs for RHA un-encapsulated die. The following is a list of possible PINs for un-encapsulated die covered by this specification sheet.

PINs for types JANHC and JANKC (1)		
JANHCA#1N821	JANHCB#1N821	JANHCD#1N821
JANHCA#1N823	JANHCB#1N823	JANHCD#1N823
JANHCA#1N825	JANHCB#1N825	JANHCD#1N825
JANHCA#1N827	JANHCB#1N827	JANHCD#1N827
JANHCA#1N829	JANHCB#1N829	JANHCD#1N829
JANKCA#1N821	JANKCB#1N821	JANKCD#1N821
JANKCA#1N823	JANKCB#1N823	JANKCD#1N823
JANKCA#1N825	JANKCB#1N825	JANKCD#1N825
JANKCA#1N827	JANKCB#1N827	JANKCD#1N827
JANKCA#1N829	JANKCB#1N829	JANKCD#1N829

(1) The number sign (#) represents one of seven RHA designators available on this specification sheet ("M", "D", "L", "R", "F", "G", or "H").

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6.6 Substitution information.

6.6.1 Substitution of radiation hardened devices. See MIL-PRF-19500.

6.6.2 Substitutability of dash-one (-1) devices. The non-dash-one devices that were in previous revisions of this document have been deleted from this specification as of MIL PR 19500/159J, dated 1 July 1998. The dash-one devices are a direct substitute for non dash-one devices and are preferred.

6.6.3 Substitution of ΔV_Z devices. The device types within this series with higher second number symbols (lower ΔV_Z voltage-temperature stability) are a direct one-way substitution for lower second number symbols (higher ΔV_Z voltage-temperature stability).

6.6.4 Superseded PINs. The following supersession data applies to PINs associated with this document:

Superseding PIN as specified within MIL-PRF-19500/159K, dated 29 January 2000	Superseded PIN as specified within MIL-PRF-19500/159J 1 July 1998
1N821-1	1N821
1N823-1	1N823
1N825-1	1N825
1N827-1	1N827
1N829-1	1N829

6.7 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCAM1N821) will be identified on the QML.

JANC ordering information						
Manufacturer						
43611 (1)		12954 (1)		12954 (1)	52GC4 (1)	
JANHCA1N821	JANHCA#1N821	JANHCB1N821	JANHCB#1N821	JANHCC1N821	JANHCD1N821	JANHCD#1N821
JANKCA1N821	JANKCA#1N821	JANKCB1N821	JANKCB#1N821	JANKCC1N821	JANKCD1N821	JANKCD#1N821
JANHCA1N823	JANHCA#1N823	JANHCB1N8233	JANHCB#1N8233	JANHCC1N823	JANHCD1N823	JANHCD#1N823
JANKCA1N823	JANKCA#1N823	JANKCB1N823	JANKCB#1N823	JANKCC1N823	JANKCD1N823	JANKCD#1N823
JANHCA1N825	JANHCA#1N825	JANHCB1N825	JANHCB#1N825	JANHCC1N825	JANHCD1N825	JANHCD#1N825
JANKCA1N825	JANKCA1N825	JANKCB1N825	JANKCB#1N825	JANKCC1N825	JANKCD1N825	JANKCD#1N825
JANHCA1N827	JANHCA#1N827	JANHCB1N827	JANHCB#1N827	JANHCC1N827	JANHCD1N827	JANHCD#1N827
JANKCA1N827	JANKCA#1N827	JANKCB1N827	JANKCB1N827	JANKCC1N827	JANKCD1N827	JANKCD#1N827
JANHCA1N829	JANHCA#1N829	JANHCB1N829	JANHCB#1N829	JANHCC1N829	JANHCD1N829	JANHCD#1N829
JANKCA1N829	JANKCA1N829	JANKCB1N829	JANKCB#1N829	JANKCC1N829	JANKCD1N829	JANKCD#1N829

(1) The number sign (#) in the PIN represents one of seven RHA designators available on this specification sheet ("M", "D", "L", "R", "F", "G", or "H").

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6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693–1642 or DSN 850–6939.

6.9 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
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Air Force - 85
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Review activities:
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